

COMP1720/6720 Practice Final Exam 2018

Note

There are no answers for the “code” questions, however there is an example answer provided for the design task (Question 3).

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Time: 15 minutes reading, 120 minutes writing

Total marks: 100

Weighting: 20%

Permitted materials: none

Make sure you read each question carefully. Questions are not equally weighted, and the size of the answer box is not necessarily related to the length of the expected answer or the number of marks given for the question.

All answers must be written in the boxes provided in this booklet. You will be provided with scrap paper for working, but only the answers written in this booklet will be marked. Do not remove this booklet from the examination room. There is additional space at the end of the booklet in case the boxes provided are insufficient. If you use these extra pages, make sure you clearly label which question the answer refers to.

Greater marks will be awarded for short, concrete answers than long, vague/rambling ones. Marks may be deducted for providing information that is irrelevant to a question.

Question 1 (30 marks total)

Part 1 2.5 marks

Which transformation has **not** been applied to the following ellipse

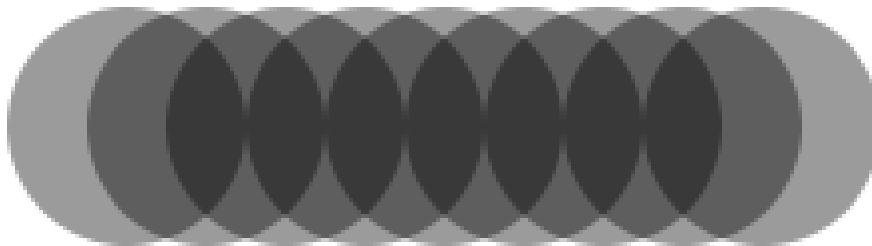
```
ellipse(0, 0, width/2, height/2);
```



- (a) `scale()`
- (b) `rotate()`
- (c) `translate()`

Part 2 2.5 marks

What `fill()` colour for the ellipses was used to draw this image?



- (a) `fill(255, 100)`
- (b) `fill(0, 100)`
- (c) `fill(100, 255)`
- (d) `fill(100)`

Part 3 2.5 marks

Which sketch (a, b or c) produced this image?



```
// sketch (a)
background(0);
fill(128);
stroke(255);
rect(width/2, height/2, width, height);
line(0, height/2, width/2, 0);

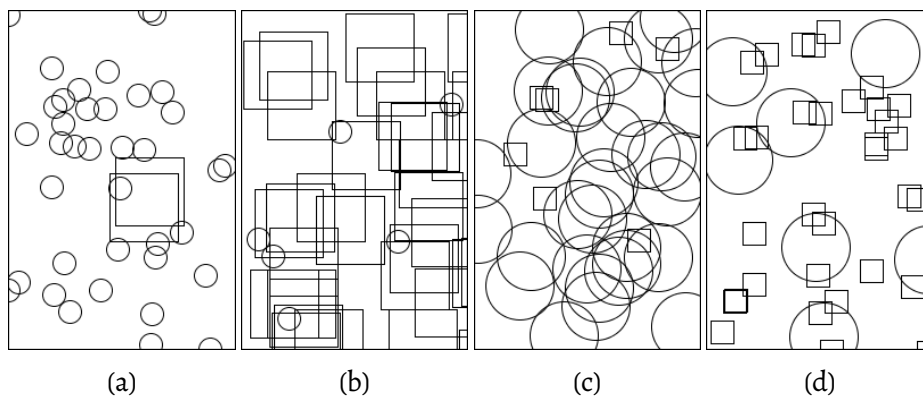
// sketch (b)
background(128);
fill(0);
rect(width/2, height/2, width, height);
stroke(255);
line(width/2, 0, height/2, 0);

// sketch (c)
background(0);
fill(128);
stroke(255);
line(0, height/2, width/2, 0);
line(width/2, height, width/2, height/2);
line(width/2, height/2, width, height/2);
```

Part 4 5 marks

Which image did this code produce?

```
for (var i = 0; i < 40; i++) {  
  if (random(100)<10)  
    rect(random(width), random(height), 20, 20);  
  else  
    ellipse(random(width), random(height), 60, 60);  
}
```

**Part 5 5 marks**

When this code runs, what gets printed to the javascript console?

```
function benFunction(x){  
  x += 15.0;  
  return x;  
}  
  
var x = 10;  
  
function draw() {  
  var x = 20 + benFunction(5);  
  console.log(x);  
  noLoop(); // code stops looping here  
}
```

(a) 35.0

(b) 25.0

(c) 40.0

(d) 10.0

Part 6 2.5 marks

```
var x = 10;
var y = -5;

x = x + y;
if(x < y)
  x = x - y;
else
  x *= 2;
```

At the end of this code, what is the value of **x**? (a) 20 (b) -5 (c) 5 (d) 10

Part 7 5 marks

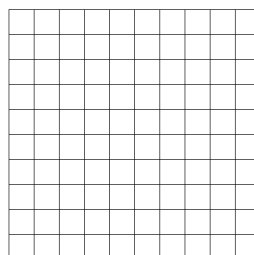
On the grid provided (1 square per pixel), draw the output of this p5 program (assuming a white background and a black stroke with `strokeWeight(1)`)

```
var arr = [[2, 5], [8, 2], [5, 8]];

function drawThing(i){
  line(i[0], 0, i[0], i[1]);
}

function setup() {
  createCanvas(10,10);
}

function draw() {
  arr.map(drawThing);
}
```



Part 8 5 marks

Describe (in words) as many similarities and differences as you can between the following code chunks.

```
// chunk 1
function doubler(v){
  return v*2;
}
function draw(){
  console.log(doubler(5));
}

// chunk 2
function doubler(v){
  console.log(v*2);
}
function draw(){
  doubler(5);
}
```

Question 2 (20 marks total)

In 2016, this exam was set about a week after the US presidential election. Your job is to create p5.js sketch which releases **red**, **white** and **blue** balloons from the ceiling (i.e. the top of the canvas).

This question has three parts. If you get stuck on any one part, you can still answer the next one—just continue as if you'd answered the previous part. Don't worry if you forget a semi-colon—you'll be marked on whether you can demonstrate that you understand the basic concepts.

Part 1 (10 marks)

Write a createBalloon function that **returns** a javascript object which:

1. contains **properties** for keeping track of the position of the balloon
2. contains **properties** for keeping track of the colour of the balloon (**red**, **white** or **blue**)
3. initialises the balloon's position to be at the **top** of the canvas
4. initialises the balloon's colour to either **red**, **white** or **blue**

Part 2 (10 marks)

Write a drawBalloon function which:

1. takes one **parameter**: an object returned from the createBalloon function
2. draws the balloon on the canvas at the correct location
3. each frame, the makes the balloon grow a little bit bigger
4. when the balloon grows too big, it “pops”

Question 3 Design task (50 marks)

Using one of the following ideas as a theme, describe a design for an original, interactive artwork in p5. Interaction with your sketch should both deepen its message and build an individual experience. Your sketch should communicate its main messages after about three minutes of interaction. You can use **text** and hand-drawn **pictures** in your response. You don't *have* to draw pictures/diagrams, but feel free to use them wherever they help you communicate the design of your sketch.

As part of your design description, address the following issues:

- what are the main ideas and/or emotions that you are trying to communicate?
- how will your envisioned interaction enhance the artistic effect of the design?
- how does your design use affordances and feedback?
- how would you realise your design using p5?

1720 themes

Note

For **COMP1720**, these are the actual themes which will be on the 2018 final exam.

If you're an *undergraduate* student in COMP1720, choose **one** theme from the following list:

1. rags to riches
2. fish out of water
3. mistaken identity

6720 themes

Note

For **COMP6720**, these are the themes from a previous year's final exam—you won't see the themes for this year's exam until you get into the examination room.

If you're a *masters* student in COMP6720, choose **one** theme from the following list:

1. under pressure
2. don't stop me now
3. crazy little thing called love

Note

You don't have to use all of the following pages for your answer—the extra pages are included in case you need them for other questions (as described on the title page).

Design task (50 marks)

Note: this is an *example* answer

It's not meant to be a the best possible answer—I'm sure you can do better than this. It's here to show you what an answer might look like for a "Swooping Magpie" theme. This theme won't be on the exam, however—for the themes on the *actual* final exam see page 9.

As always your answer to this question in the exam must **be original and be your own work**. So be careful about studying as a group. If there are a number of answers which are very similar to each other the examiners have the right to interview the students concerned, and the usual academic integrity policies & penalties apply.

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As part of your design description, address the following issues:

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Theme: Swooping Magpie

I choose the “Swooping Magpie” theme. It is a real **shock** to be swooped by a magpie. People are often swooped when riding their bicycles. But the magpies are not evil—it is just a part of nature that they feel as though they are protecting their babies. In my design, I will communicate:

- the shock of being swooped
- that magpies as a part of nature

My setting will be someone riding their bicycle. I will have an audio background of magpie calls and the creaking of a bicycle chain. I will also have some **humour** in my sketch and there will be a message about nature.

Opening scene

My opening scene will be a simple scene with someone riding a bicycle up a gentle hill. The person will be riding slowly. If you move the mouse they will ride slightly more quickly. (Actually, it is possible that moving the mouse one way will make them go forwards and the other way go backwards; I would try these ideas out when building the sketch.) You will hear magpie calls and the bicycle.

Shock! Magpie Swoop!

Suddenly, the scene will flash to black as a bird shape covers the canvas and then shrinks to be an easily-seen magpie. The magpie will zoom down and hit the rider on the head and they will fall off. The scene will then close with panels sliding forward from the left and right with the new scene drawn on them. This is like the curtains on a stage closing an act. There will be a loud squawk and a “thunk” on the audio track as the magpie hits its victim.

Accessories for rider and magpie

The next scene will be like a game where you can drag and drop items to provide accessories to the rider and the magpie. There will be large pictures of the rider (on the right panel) and the magpie (on the left panel) and a horizontal tool-bar across the bottom of the screen. The items that you can drag and drop on to the rider or the magpie are

- a big stick
- a sign saying “beware magpie”
- sun glasses
- some worms

You can only give the stick to the rider. You can only give the worms to the magpie. You can give the sign and the sun-glasses to either the rider or the magpie. You only have one choice. If you make an illegal choice, then the accessory jumps back to the tool-bar and an error message is printed out. When you have dragged the accessory to its target, then this scene closes (with closing panels as before) and the sketch returns to the riding scene again.

Note that either the rider or the magpie can wear sun-glasses! A bird wearing sun-glasses is meant to be funny. The audio track in this scene would be a “la di da” game show theme, perhaps overlaid with some magpie chirping.

Back to the swooping scene

Now we are back in the country-side watching our victim ride his/her bicycle up the hill. Except now there will be a new set of outcomes. After each outcome, this scene will close and the sketch will return to the “accessories” scene. The possible outcomes are:

- If the bicycle rider has the stick, then the magpie swoops, but does not hit the rider. But the rider still wobbles and falls off the bicycle because it is hard to ride a bicycle while holding a stick!
- If the rider or the magpie are wearing sun-glasses, then the magpie still swoops and makes the rider fall off.
- If either the rider or the magpie were given the sign, then the sign would appear in the sketch. The rider will stop to read the sign and then the magpie will swoop down and hit the rider while they are reading it.
- If the magpie was given the worms, then it will swoop down but land next to the rider and try to be the rider’s friend. But then another magpie will swoop down and hit that magpie and knock it over. (The message here is that we should leave nature alone.)

In this sketch, interaction makes it playful

In addition, the user gets to imagine what the outcomes of the different accessories might be. Interaction **slows down** the message of this sketch. In fact, it is likely that the message with the worms will be something of a punch-line for this sketch and, if so, it could have considerable impact. Of course, if the user does not choose the worms last it will not be as much of a punch-line.

This sketch is feasible in p5.js

The biggest challenge would be constructing a good animation of the bicycle riding and the rider falling off. I would use cartoon images for the rider and a simple schematic for the bicycle; but this will be challenging. The magpie would be an image with various amounts of dynamic scaling to shrink it down and hit the rider. The audio track would be easy to construct from field recordings or to come by from the internet.